

A

REAL-TIME RESEARCH PROJECT REPORT ON

## **SMART SHOPPING CART**

Submitted in partial fulfillment of the requirement for second year  
of

**BACHELOR OF TECHNOLOGY**

IN

**INFORMATION TECHNOLOGY**

*Submitted by*

**V. AKHILAN (23P61A12C4)**

**M. PRUTHVIK (22P61A1275)**

**R. UDAY (23P61A1295)**

Under the Supervision

*of*

**Mrs. T. Sreelalitha**

Assistant Professor, Department of Information Technology



**VIGNANA  
BHARATHI**

Department of Information Technology

**VIGNANA BHARATHI INSTITUTE OF TECHNOLOGY**

(Approved by AICTE, Accredited by NBA, NAAC, Permanently Affiliated to  
JNTUH) Aushapur (v), Ghatkesar (m), Medchal.dist, TELANGANA-501 301

JUNE-2025



AUSHAPUR(V), GHATKESAR(M), MEDCHAL.DIST-501301

**Department of Information Technology**

## **CERTIFICATE**

This is to certify that the project entitled **“SMART SHOPPING CART”** is being submitted by **V. AKHILAN (23P61A12C4), M. PRUTHVIK (22P61A1275), R. UDAY (23P61A1295)** in partial fulfillment of the requirement for second year of **Bachelor of Technology in Information Technology** is a record of bonafide work carried out by them under my guidance and supervision during the academic year 2024-2025. The results embodied in this project report have not been submitted to any other University for the award of any degree or diploma.

**Supervisor**

**Mrs. T. Sreelalitha**

**Asst. Professor**

**Department of IT**

**Project coordinator**

**Mr. P. Shiva Kumar**

**Asst. Professor**

**Department of IT**

**Head of the**

**department**

**Dr K. Kalaivani**

**Head of the**

**Department of IT**



AUSHAPUR(V), GHATKESAR(M), MEDCHAL. DIST-501301

## **Department of Information Technology**

### **DECLARATION**

We, **V. AKHILAN** bearing hall ticket number **23P61A12C4**, **M. PRUTHVIK** bearing hall ticket number **22P61A1275** and **R. UDAY** bearing hall ticket number **23P61A1295**, hereby declare that the project report entitled “**SMART SHOPPING CART**” under the guidance of **Mrs. T. Sreelalitha**, Department of Information Technology, **VBIT**, Hyderabad, is submitted in partial fulfillment of the requirement for second year of Bachelor of Technology in Information Technology.

This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other university or institute for the award of any other degree or diploma.

**V. AKHILAN (23P61A12C4)**

**M. PRUTHVIK (22P61A1275)**

**R. UDAY (23P61A1295)**

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First and foremost, we wish to express our gratitude to our **Supervisor, Mrs. T. Sreelalitha**, Asst. Professor, Department of Information Technology for her valuable guidance and freedom she gave to us.

We also express our sincere thanks to **Mr. P. Shiva Kumar & Ms. P. Sony, Project Coordinator** for their encouragement and support throughout the project.

We also express our sincere thanks to **Dr K. Kalaivani, HOD of IT** for her encouragement and support throughout the project.

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Our utmost thanks also go to all the **FACULTY MEMBERS** and **NON-TEACHING STAFF** of the Department of Information Technology for their support throughout our project work.

**V. AKHILAN (23P61A12C4)**

**M. PRUTHVIK (22P61A1275)**

**R. UDAY (23P61A1295)**

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# VIGNANA BHARATHI INSTITUTE OF TECHNOLOGY

## Department of Information Technology

### COURSE OUTCOMES

**Course:** Real Time or Field Based Research Project Course(22IT2281)

**Class:** II B Tech II Semester

**AY:** 2024-2025

#### Course Outcomes

After completing the Projects, the student will be able to:

| Code | Course Outcomes                                                                                | Taxonomy |
|------|------------------------------------------------------------------------------------------------|----------|
| C229 | Identify and state the problem precisely to prepare the abstract                               | Remember |
| C229 | Analyze the existing system, and outlining the proposed methodology for effective solution     | Analyze  |
| C229 | Use various modern tools for designing applications based on specified requirements            | Apply    |
| C229 | Develop applications with adequate features and evaluate the application to ensure the quality | Create   |
| C229 | Prepare the document of the project as per the guidelines                                      | Create   |

### PROGRAM OUTCOMES (POs)

- 1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- 2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- 3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- 4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- 6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- 7. Environment and sustainability:** Understand the impact of the professional engineering solutions

in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

**8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

**9. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

**10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

**11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

**12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

### **PROGRAM SPECIFIC OUTCOMES(PSO's)**

**PSO1** Simulate computer hardware and apply software engineering principles and techniques to develop various IT applications

**PSO2** Analyze various networking concepts and also aware of how security policies, standards and practices are used for trouble-shooting.

**PSO3** Design and maintain database for providing back-end support to software projects.

**PSO4** Apply algorithms and programming paradigms to produce IT based solutions for the real- world problems.

# VIGNANA BHARATHI INSTITUTE OF TECHNOLOGY

## Department of Information Technology

### COs Mapping with PO/PSO

**Project Title:** Smart Shopping Cart

**Name of the Supervisor:** Mrs. T. Sreelalitha

**Batch Details:**

| S. No | ROLLNO     | STUDENT NAME | TECHNOLOGY |
|-------|------------|--------------|------------|
| 1     | 23P61A12C4 | V. AKHILAN   | IOT        |
| 2     | 22P61A1275 | M. PRUTHVIK  | IOT        |
| 3     | 23P61A1295 | R. UDAY      | IOT        |

### CO-PO Mapping for Major Project:

High-3 Medium-2 Low-1

| PO/<br>C O | PO<br>1 | PO<br>2 | PO<br>3 | PO<br>4 | PO<br>5 | PO<br>6 | PO<br>7 | P08 | PO<br>9 | PO<br>10 | PO<br>11 | PO<br>12 | PS<br>O1 | PS<br>O2 | PS<br>O3 | PS0<br>4 |
|------------|---------|---------|---------|---------|---------|---------|---------|-----|---------|----------|----------|----------|----------|----------|----------|----------|
| C229.1     | 3       | 3       | 2       | 3       | 3       | 2       | -       | -   | 3       | 2        | 3        | 3        | 2        | -        | -        | -        |
| C229.2     | 2       | 2       | 3       | 2       | 3       | 2       | -       | -   | 3       | 2        | 3        | 3        | 2        | -        | -        | -        |
| C229.3     | 2       | 2       | 3       | 2       | 3       | 2       | -       | -   | 3       | 2        | 3        | 3        | 2        | -        | -        | -        |
| C229.4     | 2       | 2       | 3       | 2       | 3       | 2       | -       | -   | 3       | 2        | 3        | 3        | 2        | -        | -        | -        |
| C229.5     | 2       | 2       | 2       | 2       | 3       | 2       | -       | -   | 3       | 2        | 3        | 3        | 2        | -        | -        | -        |
| AVG        | 2.2     | 2.2     | 2.6     | 2.2     | 3       | 2       | -       | -   | 3       | 2        | 3        | 3        | 2        | -        | -        | -        |

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## Abbreviations and Acronyms

| SL.NO | ACRONYM | EXPANSION                          |
|-------|---------|------------------------------------|
| 1     | IOT     | Internet of Things                 |
| 2     | UID     | Unique Identifiers                 |
| 3     | LCD     | Liquid Crystal Display             |
| 4     | RFID    | Radio Frequency Identification     |
| 5     | GPIO    | General Purpose Input-Output       |
| 6     | IDE     | Integrated Development Environment |
| 7     | ICSP    | In-Circuit Serial Programming      |
| 8     | MCU'S   | Microcontroller Unit               |
| 9     | USB     | Universal Serial Bus               |

## ABSTRACT

Shopping for groceries and daily items is something everyone does. But often, people face problems like long lines at the billing counter, not knowing the total cost of their items while shopping, and spending extra time looking for product information. These issues make shopping slow and stressful. At the same time, shop owners want to make the shopping process faster and reduce the number of staff needed for billing and support.

To solve these problems, we are creating a **Smart Shopping Cart** using RFID (Radio Frequency Identification) technology. In this system, every product in the store will have a small RFID tag attached to it. The shopping cart will have an RFID reader, an Arduino board (small computer), and an LCD screen.

When a customer puts an item into the cart, the RFID reader will automatically read the tag and show the item's name, price, and total cost on the screen. If the customer removes the item from the cart, the display will update the cost and quantity again. This way, the customer always knows what they have added and how much it costs—without needing to wait in line for billing.

This smart cart will help customers save time and shop more comfortably. It will also help store owners by reducing the need for extra staff and improving the overall shopping experience. With this system, shopping becomes quicker, smarter, and more convenient.

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# 1.INTRODUCTION

## 1.1 INTRODUCTION

Internet of Things (IOT) is a concept where each device is assigned to an IP address and through that IP address anyone makes that device identifiable on internet. The mechanical and digital machines are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction. Basically, it started as the “Internet of Computers.” Research studies have forecast an explosive growth in the number of “things” or devices that will be connected to the Internet. The resulting network is called the “Internet of Things” (IoT). The recent developments in technology which permit the use of wireless controlling environments like, Bluetooth and Wi-Fi that have enabled different devices to have capabilities of connecting with each other. Using a WIFI shield to act as a Micro web server for the Arduino which eliminates the need for wired connections between the Arduino board and computer which reduces cost and enables it to work as a standalone device.

## 1.2 BACKGROUND

Shopping in supermarkets can be time-consuming and stressful. Customers often have to wait in long queues, worry about the total cost of their items, and get little help with product information. At the same time, store owners face high labor costs and customer dissatisfaction.

With the rise of new technologies like **RFID** and **IoT**, shopping can be made smarter and faster. A smart shopping cart can help solve these problems by automatically reading items, showing their price, and updating the total cost in real time. This makes shopping more convenient for customers and reduces manual work for store staff.

## 1.3 PROJECT OBJECTIVES

The objective of this project is to design and implement a Smart Shopping Cart using an Arduino Nano, which makes shopping easy for customers.

The system aims to:

- To automate the billing process by using RFID technology to detect items added or removed from the cart.
- To display item details in real-time, including name, quantity, and price, on an LCD screen.
- To reduce customer waiting time at the checkout counter by eliminating manual billing.
- To provide a user-friendly shopping experience with instant cost updates and item tracking.
- To help store owners reduce labor costs by minimizing the need for billing staff.
- To improve inventory management by keeping track of items added to the cart automatically.
- To implement a cost-effective solution using readily available hardware like Arduino and RFID modules.

## 1.4 SCOPE

The scope of this project is to design and develop a Smart Shopping Cart system that enhances the in-store shopping experience using RFID technology. The system will be implemented in supermarkets, grocery stores, and retail outlets to automate item detection and billing. Each product will have an RFID tag, and the shopping cart will be equipped with an RFID reader, Arduino board, and LCD display. As customers add or remove items from the cart, the system will automatically detect the changes and display the item names, prices, and total cost in real time. This will help customers track their purchases and manage their spending without the need to wait in long billing queues. The project aims to reduce labor costs for store owners and improve shopping convenience for users. This version of the system will focus on in-cart automation and display features and does not include online payment or advanced inventory management. However, it serves as a prototype that can be expanded in the future with additional functionalities.

## 1.5 PROJECT MANAGEMENT

Management of any project can be briefly disintegrated into several phases. Our project has been decomposed into the following phases:

### Experimentation

This phase involved discussions regarding necessary equipment regarding the project. The study of related already existing projects, gathering required theoretical learning. It also included figuring out the coding part, by developing simple algorithms and flowcharts to design the whole process.



**Fig .5.1.1: Model of Phases in Project Management**

### Design

This phase was, designing layout of the application, and the necessary features to be included. This involved the complete hardware assembly and installing the code to Arduino Nano. The circuit diagram and hardware layout were drawn, and all components were integrated accordingly.

### Development and testing

This stage included the actual coding and uploading of the program onto the Arduino Nano. The RFID reader was tested to ensure it could successfully detect different RFID tags attached to products. The LCD display was also tested for correct item name, price, quantity, and total cost output.

### Real world testing

The prototype was ready to be tested into the real world and integrated with various real time electrical appliances.

## 1.6 OVERVIEW AND BENEFITS

- **Real-Time Billing and Price Display**  
Automatically detects items added or removed from the cart and updates the total price instantly, helping customers keep track of their expenses during shopping.
- **Enhanced Shopping Experience**  
Reduces waiting time at checkout counters by enabling automatic billing and providing customers with real-time information about their purchases.
- **Labor Cost Reduction**  
Minimizes the need for billing staff and cashiers, allowing store owners to save on labor costs and improve operational efficiency.
- **Ease of Use**  
The system uses an easy-to-read LCD display to show item details and total cost, making it simple for customers of all ages to use.
- **Scalability and Flexibility**  
Can be expanded to include features like mobile app integration, automatic payment, or inventory management with minimal hardware changes.
- **Improved Inventory Tracking**  
Automatically records the items picked by customers, helping store owners maintain better control over stock and reduce theft or errors.

## 1.7 ORGANIZATION OF THESIS

**Chapter 1** The thesis is organized into seven chapters including the introduction. Each chapter is unique on its own and is described with necessary theory to comprehend it.

**Chapter 2** deals with *Literature Review*, this chapter reflects a comprehended form of the existing projects related to the topic. It credits the projects along with a brief paragraph of summary about the project.

**Chapter 3** describes the *Hardware Modelling and setup* of the project. The chapter points the main features of the prototype, gives a layout of the project, lists the components required. It briefly describes the various setup processes involved with the project, including hardware interfacing and software installation and setup according to our requirement.

**Chapter 4** is the *Logic and operation* of the project. A flow chart presents the actions describing the working process of the prototype.

**Chapter 5** is the *Result*. It describes the results of the proposed system.

**Chapter 6** is the *conclusion and Future scope*. This chapter includes the result of the project work carried, the limitations it possesses.

**Chapter 7** is the *Future enhancements*. It describes the Future Enhancements of the proposed system.

**Chapter 8** lists the *References* that have been used for the commencement of the project work.

## 2.LITERATURE REVIEW

**2.1 Title: Smart Shopping Cart Using RFID Technology**  
**Journal: International Journal of Computer Applications (IJCA)**  
**Authors: S. Kumar, A. Gupta, M. Sharma**

This paper presents a smart shopping cart system that uses RFID tags on products and an RFID reader on the cart to automatically detect items as they are added or removed. The system updates the total price in real-time and displays it on an LCD screen. The paper also discusses challenges such as tag reading accuracy and system cost.

**2.2 Title: Automated Billing System Using RFID in Retail Stores**  
**Journal: International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering (IJIREEICE)**  
**Authors: R. Patel, K. Joshi, N. Desai**

This study describes an RFID-based automated billing system for retail applications. Using Arduino microcontrollers and RFID readers, the system identifies products instantly and calculates the bill without manual intervention. The research highlights cost reduction in labor and increased accuracy in billing. It also explores integration with mobile apps for remote monitoring.

**2.3 Title: Implementation of a Smart Cart System Using Arduino and RFID**  
**Journal: International Journal of Advanced Research in Computer Engineering & Technology (IJARCET)**

**Authors: A. Singh, P. Verma, D. Kumar**

This paper details the design and development of a smart shopping cart prototype that uses Arduino and RFID to track shopping items. The cart displays real-time product information and total cost to the customer via an LCD display. The system aims to enhance shopping efficiency and customer satisfaction. The authors emphasize the ease of use and potential for scalability.

**2.4 Title: RFID-Based Smart Cart for Supermarket Automation**  
**Journal: International Journal of Engineering and Technology (IJET)**  
**Authors: M. Das, S. Roy, P. Chatterjee**

This research focuses on an RFID-based smart cart that automatically scans items placed inside and updates the bill instantly. The system integrates hardware like Arduino and RFID readers and software for item tracking. The authors discuss the reduction of queue times and labor costs and present experimental results from prototype testing.

**2.5 Title: Review on Smart Shopping Cart System Using RFID and IoT**  
**Journal: International Journal of Engineering Research & Technology (IJERT)**  
**Authors: V. Sharma, R. Singh, H. Mehta**

This survey paper reviews various smart shopping cart solutions using RFID and IoT technologies. It compares different hardware setups, communication protocols, and display methods. The authors analyze the benefits of real-time billing, inventory management, and customer convenience. Challenges such as data security and tag interference are also discussed, along with suggestions for future improvements.

## 3.HARDWARE MODELLING AND SETUP

### 3.1 MAIN FEATURES OF THE PROTOTYPE

- **Automatic Item Detection**  
Uses an RFID reader mounted on the cart to detect RFID tags attached to products as they are added or removed.
- **Real-Time Price and Quantity Display**  
An LCD screen shows the names, quantities, and prices of items in the cart, updating instantly with each item scanned.
- **Arduino-Based Control**  
An Arduino microcontroller processes RFID data and manages the display, enabling seamless communication between hardware components.
- **User-Friendly Interface**  
Clear and simple LCD messages provide customers with real-time feedback on the items and total cost in their cart.
- **Fully Automated Operation**  
The system works without manual input for billing, reducing human error and speeding up the shopping process.
- **Low-Cost and Energy Efficient**  
Built with affordable and readily available components like Arduino, RFID modules, and LCDs, ensuring low power consumption and cost-effectiveness.

### 3.2 PROJECT LAYOUT

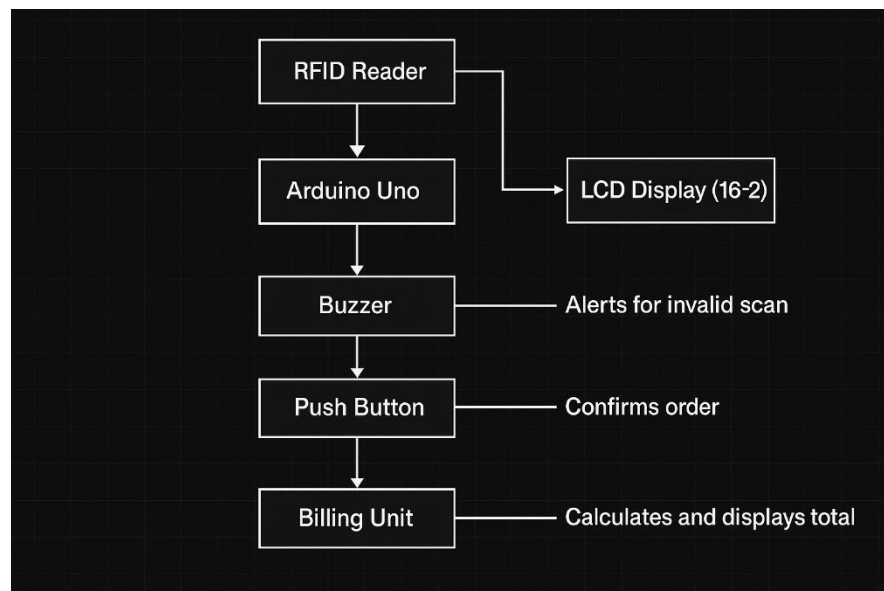


Fig 3.2.1: Layout of Project Module

- **Arduino Nano:** Arduino Nano is a low-cost, flexible, and easy-to-use programmable open-source microcontroller board that can be integrated into various electronic circuits.
- **LCD Display:** An LCD consists of a layer of liquid crystals sandwiched between two transparent electrodes. When an electric current is applied, the crystals align to control the amount of light passing through them, creating the image you see on the screen.
- **Bread Board:** A breadboard (sometimes called protoboard) is essentially the foundation to

construct and prototype electronics. A breadboard allows for easy and quick creation of temporary electronic circuits or to carry out experiments with circuit design.

- **IR Sensors:** IR Sensor 1 detects vehicles approaching the entrance to trigger gate control, while IR Sensor 2 detects whether the parking slot is occupied. Both sensors send real-time signals to the Arduino for automated decision- making.
- **NANO ports:** All Arduino ports have inbuilt ports but if not available will be available when Arduino USB port is inserted.

### 3.3 COMPONENTS REQUIRED

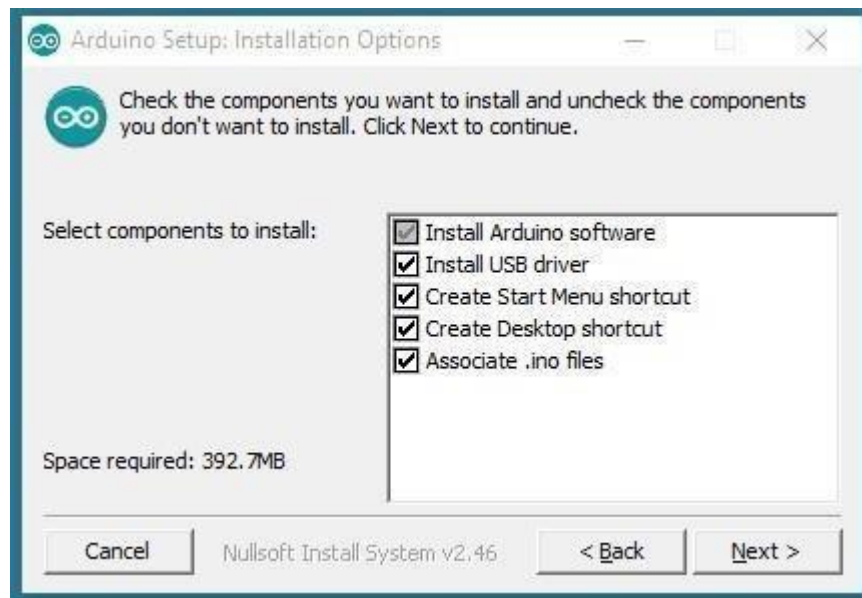
| S.NO | COMPONENT     | QUANTITY |
|------|---------------|----------|
| 1.   | Arduino Nano  | 1        |
| 2.   | Bread Board   | 1        |
| 3.   | LCD Display   | 1        |
| 4.   | Jumper Wires  | 15       |
| 5.   | Servo Motor   | 1        |
| 6.   | RFID Scanner  | 1        |
| 7.   | RFID Tags     | 5        |
| 8.   | I2C Interface | 1        |
| 9.   | USB Cable     | 1        |
| 10.  | Push Button   | 1        |

**Table 3.3.1: Component List**

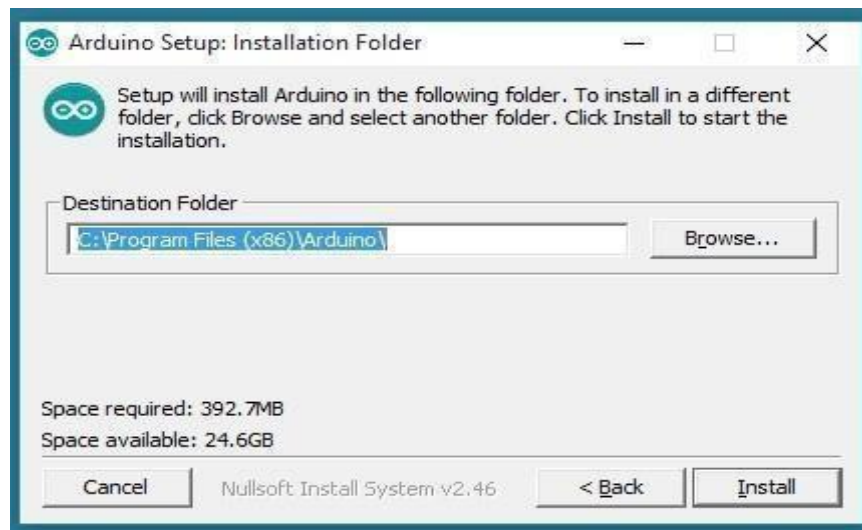
### 3.4 SETTING UP THE SYSTEM

#### 3.4.1 INSTALLATION OF ARDUINO IDE

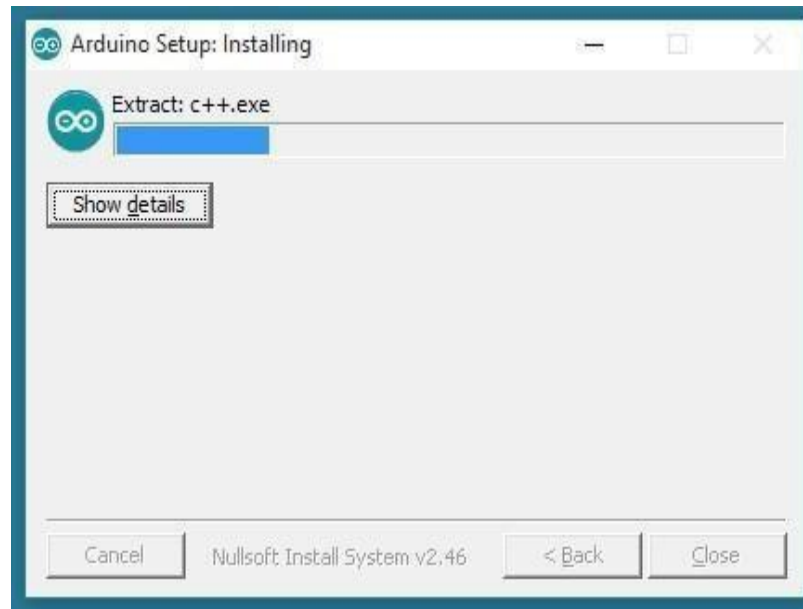
Get the latest version from the [download page](#). You can choose between the Installer (.exe) and the Zip packages. We suggest you use the first one that installs directly everything you need to use the Arduino Software (IDE), including the drivers.



**Fig 3.4.2: Choose Components to download Arduino IDE**



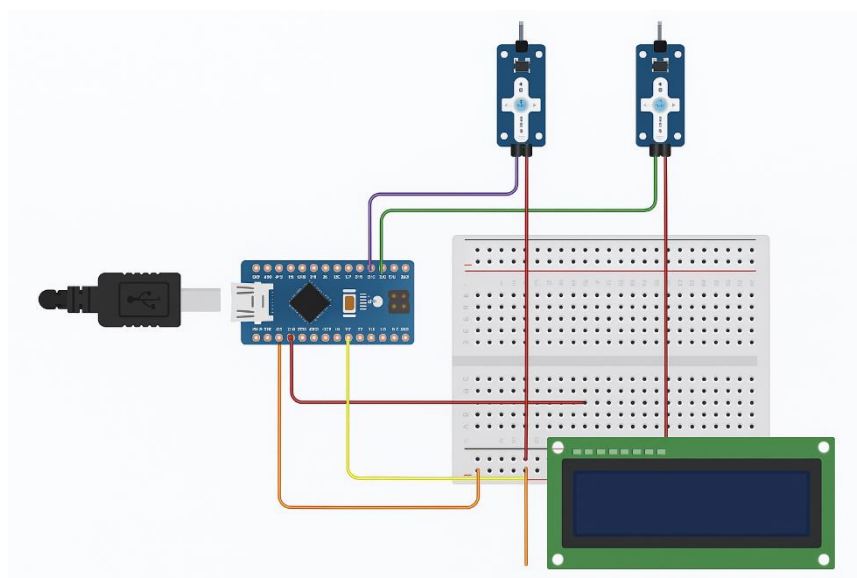
**Fig 3.4.3: Choose Arduino Installation directory**



**Fig 3.4.4: Installation of Arduino IDE**

### 3.5 HARDWARE ASSEMBLY

The Arduino NANO is a standard board of Arduino. Here NANO means 'one' in Italian. It was named as NANO to label the first release of Arduino Software. It was also the first USB board released by Arduino. It is considered as the powerful board used in various projects. Arduino.cc developed the Arduino NANO board.



**Fig 3.5.1: Pin Configuration of Arduino Nano**

Arduino Nano is based on an ATmega Microcontroller. It is easy to use compared to other boards, such as the Arduino Mega board, etc. The board consists of digital and analog Input/Output pins (I/O), shields, and other circuits. The Arduino NANO includes 6 analog pin inputs, 14 digital pins, a USB connector, a power jack, and an ICSP (In-Circuit Serial Programming) header. It is programmed based on IDE, which stands for Integrated Development Environment. It can run on

both online and offline platforms. The IDE is common to all available boards of Arduino.

**ICSP pin** - The In-Circuit Serial Programming pin allows the user to program using the firmware of the Arduino board.

**Power LED Indicator**- The ON status of LED shows the power is activated. When the power is OFF, the LED will not light up.

**Digital I/O pins**- The digital pins have the value HIGH or LOW. The pins numbered from D0 to D13 are digital pins.

**TX and RX LED's**- The successful flow of data is represented by the lighting of these LED's.

**AREF**- The Analog Reference (AREF) pin is used to feed a reference voltage to the Arduino NANO board from the external power supply.

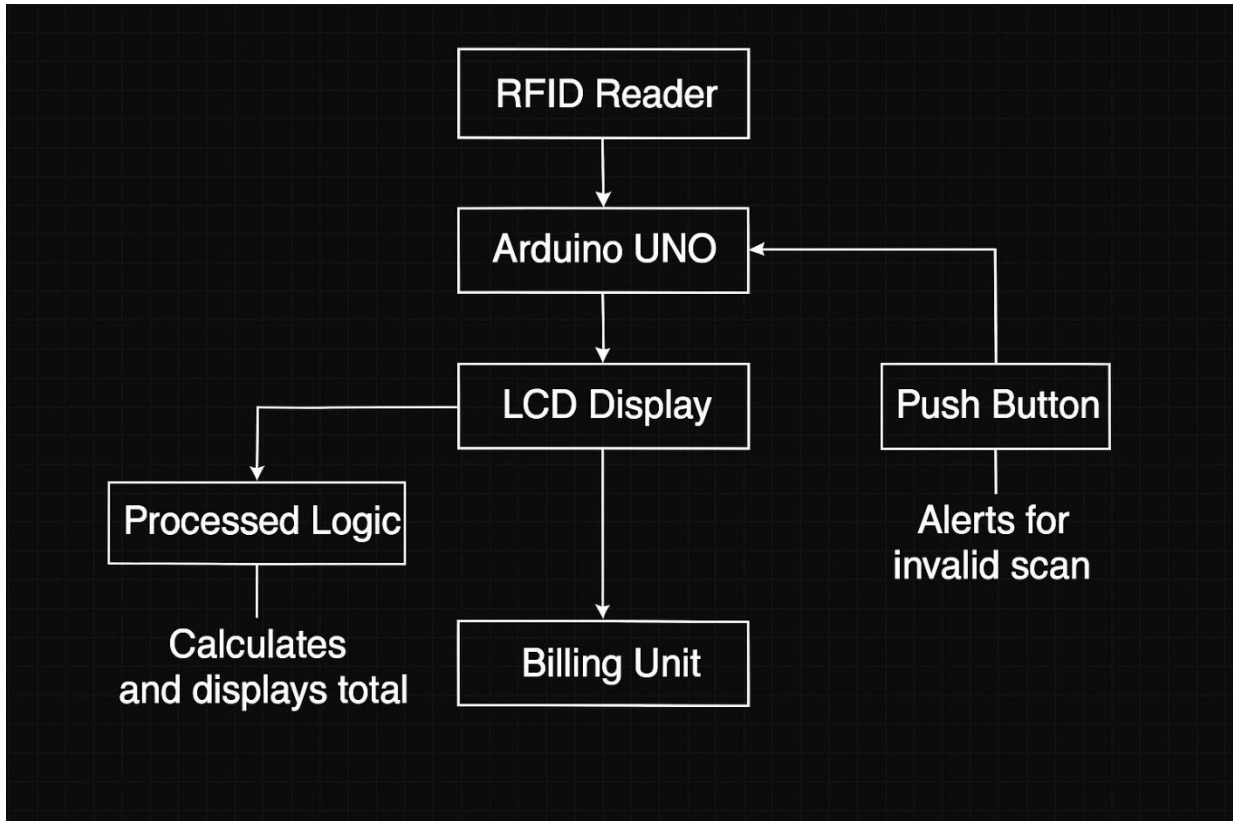
**Reset button**- It is used to add a Reset button to the connection.

**USB**- It allows the board to connect to the computer. It is essential for the programming of the Arduino NANO board.

## 4.LOGIC AND OPERATION

### 4.1 BLOCK DIAGRAM

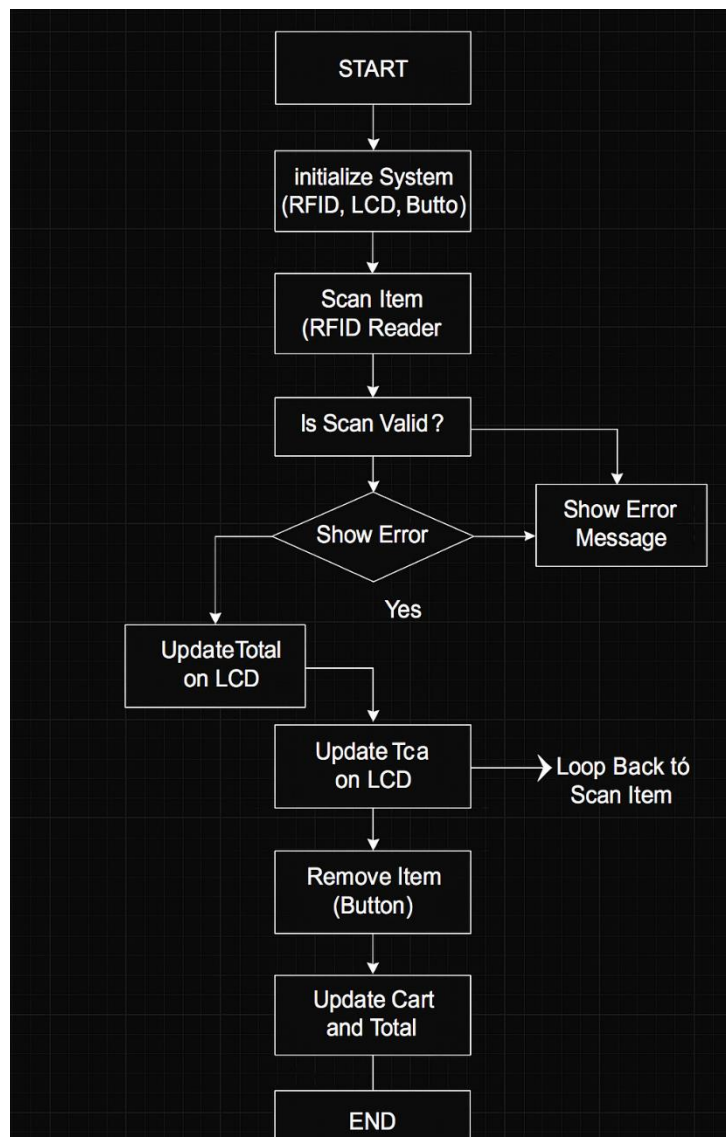
#### 4.1 Block Diagram of Proposed System



**Fig 4.1.1: Block Diagram of Proposed System**

This block diagram provides a high-level overview of the components and their interactions in a Smart Shopping Cart system using Arduino. Depending on specific project requirements, additional modules may be integrated, such as RFID readers for user identification, Bluetooth/Wi-Fi modules for data transfer, or weight sensors for precise billing and inventory control.

## 4.2 FLOW CHART



**Fig 4.2.1: Flow chart of the proposed project**

This flowchart illustrates the working of the proposed smart shopping cart system. When the push button is pressed (indicating an item is added or an action is taken), the input is read by the Arduino Uno. The Arduino processes this input and updates the item details or total cost, which is then displayed on the LCD screen. Based on this interaction, further actions—such as triggering a buzzer, logging data, or transmitting information—can be managed by the microcontroller to enhance the user's shopping experience.

## 4.3 PRINCIPLE AND OPERATION

The Smart Shopping Cart system using Arduino Nano operates on the principle of automated product identification and user interaction to facilitate efficient shopping. The cart is equipped with an RFID scanner that detects RFID tags attached to products as they are placed inside the cart. Each detected item's information is sent to the Arduino Nano, which processes and displays the item details and running total price on the LCD display via the I2C interface.

A push button is included to allow the user to finalize their selection or remove items, updating the cart's content accordingly. The servo motor is employed to control a small lock mechanism or alert signal on the cart, activated when checkout is complete or if unauthorized removal of items is detected.

The Arduino Nano connects all components through jumper wires and a breadboard for prototyping, powered and programmed through a USB cable. This system automates the shopping process by reducing manual item scanning, providing real-time billing information, and enhancing convenience for both shoppers and store management.

### 4.3.1 ADVANTAGES OF ARDUINO NANO

#### **Easy to Program**

Arduino Nano uses a beginner-friendly programming environment (Arduino IDE), making it easy to write, upload, and modify code for shopping cart logic.

#### **Cost-Effective**

It is an affordable microcontroller, which makes it ideal for low-cost smart shopping solutions especially for small-scale or prototype projects.

#### **Sufficient I/O Pins**

With 14 digital and 6 analog input/output pins, Arduino Nano can easily handle multiple sensors (like RFID) and output devices (like LCD and servo motors) in a smart shopping cart.

#### **Compact and Portable**

Its small size and lightweight design make it suitable for integration of cart.

#### **Wide Community Support**

A large online community and extensive documentation ensure easy troubleshooting, updates, and upgrades, which is beneficial for student or DIY projects.

#### **Real-Time Processing**

Arduino Nano can process sensor input and control actuators like servo motors in real-time.

#### **Expandability**

The board supports shields and modules (like GSM, Bluetooth, or Wi-Fi), allowing future upgrades like mobile notifications or QR scanning.

#### **Low Power Consumption**

Arduino Nano consumes minimal power, which is ideal for systems designed to run continuously.

### 4.3.2 DISADVANTAGES OF ARDUINO NANO

#### **Limited Processing Power and Memory**

The Arduino Nano has a relatively low processing power and limited memory compared to more advanced microcontrollers.

#### **Limited Number of I/O Pins**

The Arduino Nano provides a limited number of input/output pins (14 digital pins, 6 of which can be used for PWM outputs, and 6 analog input pins). This can be a constraint if the system needs to

interface with multiple sensors, actuators, displays, or communication modules simultaneously.

### **No Built-In Real-Time Clock (RTC)**

The Arduino Nano does not have a built-in real-time clock (RTC). For applications requiring precise timekeeping an external RTC module is necessary.

### **Limited Connectivity Options**

While the Arduino Nano can be paired with connectivity modules like Bluetooth (e.g., HC-05) or WiFi (e.g., ESP8266), it lacks built-in connectivity options. This means additional hardware and complexity are required for remote monitoring and control. Integrating these modules can also be challenging for beginners.

### **Power Consumption**

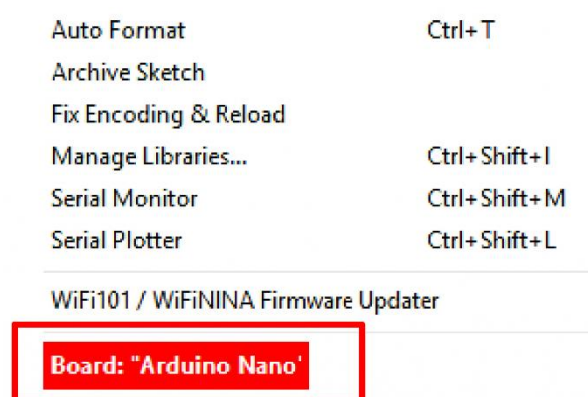
Although the Arduino Nano is relatively energy-efficient, it may still consume more power than some dedicated temperature control ICs or simpler microcontroller units (MCUs).

## **4.4 APPLICATION**

### **4.4.1 INTERFACING ARDUINO NANO WITH ARDUINO IDE**

1. Click on Start -> Control Panel -> System and Security.
2. Click on System -> Device Manager -> Ports (COM & LPT) -> Arduino NANO (COMxx). If the COM & LPT is absent, look at Other Devices -> Unknown Device.
3. Right-click to Arduino NANO (COMxx) -> Update Driver Software -> Browse my computer for driver software.
4. Select the file "inf" to navigate else, select "ArduinoNANO.inf" .
5. Installation Finished.

Open the code or sketch written in the Arduino software. Select the type of board. Click on 'Tools and select the Board.



**Fig 4.4.1.1: Selection of Arduino Nano in Arduino IDE**

- Select the port. Click on the Tools -> Port (select the port). The port likely will be COM3 or higher. For example, COM6, etc. The COM1 and COM2 ports will not appear, because these

two ports are reserved for the hardware serial ports.

- Now, upload and run the written code or sketch.

Upload and run, click on the button present on the top panel of the Arduino display:



**Fig 4.4.1.2: Uploading the Code**

Within the few seconds after the compile and run of code or sketch, the RX and TX light present on the Arduino board will flash. The 'Done Uploading' message will appear after the code is successfully.

#### **4.5 COST ESTIMATION**

| <b>S.NO</b> | <b>COMPONENTS</b> | <b>QUANTITY</b> | <b>PRICE</b> |
|-------------|-------------------|-----------------|--------------|
| 1.          | Arduino Nano      | 1               | 400          |
| 2.          | Bread Board       | 1               | 60           |
| 3.          | LCD Display       | 1               | 125          |
| 4.          | Jumper Wires      | `15             | 50           |
| 5.          | Servo Motor       | 1               | 100          |
| 6.          | RFID Scanner      | 1               | 500          |
| 7.          | RFID Tags         | 5               | 75           |
| 8.          | I2C Interface     | 1               | 75           |
| 9.          | USB Cable         | 1               | 50           |
| 10.         | Push Button       | 1               | 10           |
| Total       |                   |                 | 1445         |

**Table 4.5.1: Cost Estimation**

## 5.RESULT

### 5.1 RESULT

The Smart Shopping Cart developed using Arduino Nano, an RFID scanner, RFID tags, a servo motor, an LCD display with I2C interface, a push button, and jumper wires successfully automates the process of item detection and billing during shopping. The system accurately reads RFID-tagged products placed inside the cart and displays item details and the total price in real-time on the LCD. When the push button is pressed, the Arduino finalizes or resets the cart contents based on the input.

The servo motor can be configured to lock the cart or trigger a signal once the checkout is complete, enhancing the security of the system. The LCD provides clear and timely feedback to the user, making the cart interactive and user-friendly. The system operates smoothly, demonstrating fast response times and reliable data processing, all within a low-cost and modular hardware setup.

This project reduces dependency on manual billing, shortens checkout times, and improves shopping efficiency. It serves as a practical and scalable prototype for future smart retail applications and can be especially useful in supermarkets, grocery stores, or departmental outlets aiming to enhance customer experience through automation.



**Fig 5.1.1: Result**

## **6.CONCLUSION**

### **6.1 CONCLUSION**

The Smart Shopping Cart using Arduino Nano effectively demonstrates how automation and embedded systems can revolutionize the traditional shopping experience. By integrating components such as an RFID scanner, LCD display, servo motor, and push button, this project presents a functional prototype capable of real-time product identification, price calculation, and checkout assistance. The Arduino Nano serves as the central controller, ensuring accurate data processing and smooth interaction between the hardware components.

The system successfully detects products using RFID tags as they are placed in the cart, displaying item information and running totals on the LCD via the I2C interface. The push button allows the user to confirm or reset their cart, while the servo motor can be used to trigger a lock or alert mechanism to enhance cart security and prevent theft or item mismatch. The use of basic yet effective components like jumper wires and a breadboard makes the system modular, cost-effective, and ideal for prototyping or educational use.

This project addresses key retail challenges such as long checkout queues and manual billing errors by introducing an automated, user-friendly approach to shopping. It reduces human effort, enhances accuracy, and saves time for both customers and store staff. The implementation is especially suited for supermarkets, departmental stores, and self-service retail outlets.

Moreover, the Smart Shopping Cart system offers great potential for future development. Enhancements such as wireless payment integration, Bluetooth or Wi-Fi connectivity for syncing with store databases, mobile app support, and inventory tracking can significantly boost the system's capabilities. These upgrades would not only optimize the retail process but also align the system with the vision of smart retail environments and IoT-based commerce.

In conclusion, the Smart Shopping Cart using Arduino Nano is an innovative and efficient solution for modernizing the shopping experience. It combines real-time automation, affordability, and practicality, paving the way for smarter and more interactive retail systems. The project meets its objective and serves as a strong foundation for future innovations in the field of smart retail and embedded system applications.

## **7. FURTHER ENHANCEMENT**

To improve the functionality, user experience, and scalability of the Smart Shopping Cart system, several enhancements can be implemented. One of the key upgrades would be the integration of wireless communication modules such as Bluetooth, Wi-Fi, or GSM to enable real-time syncing of product data with a central store server or cloud database. This would allow seamless inventory management, digital receipts, and remote monitoring of carts.

Developing a dedicated mobile application could significantly enhance user interaction by displaying detailed billing information, offering personalized promotions, and allowing users to track their shopping list in real-time. Implementing automatic payment features using NFC or QR code-based mobile payments could eliminate the need for traditional checkouts, further streamlining the shopping process.

For improved item detection accuracy, barcode or QR code scanning modules could be added alongside RFID, enabling support for a wider range of product tagging systems. Load cells or weight sensors could also be integrated to verify item quantities and prevent billing errors or theft. The addition of GPS modules would enable cart tracking within large stores, helping users locate items or navigate the store more efficiently.

To increase energy efficiency and make the system eco-friendly, solar-powered modules could be integrated for outdoor market carts. Additionally, voice feedback or multilingual display support could be added to assist users with visual impairments or language barriers.

By connecting the system to cloud platforms, stores could analyze customer behavior, optimize shelf arrangements, and manage stock levels more effectively. These insights would be valuable for both inventory planning and marketing strategies.

Incorporating these enhancements would transform the Smart Shopping Cart from a prototype into a fully functional, intelligent retail solution. It would not only improve the shopping experience for customers but also bring significant operational benefits to store owners, paving the way for smarter, data-driven retail environments.

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### **8.1 REFERENCES**

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# SMART SHOPPING CART

<sup>1</sup>Mrs. T. Sreelalitha  
Assistant Professor  
(IT Dept)  
VBIT, Hyderabad,  
India

<sup>2</sup> V. Akhilan  
UG Scholar  
VBIT, Hyderabad, India

<sup>3</sup>M. Pruthvik  
UG Scholar  
VBIT, Hyderabad, India

<sup>4</sup>R. Uday  
UG Scholar  
VBIT, Hyderabad, India

[akhilanvengala9@fgmail.com](mailto:akhilanvengala9@fgmail.com)

## ABSTRACT:

*Shopping for groceries and daily items is something everyone does. But often, people face problems like long lines at the billing counter, not knowing the total cost of their items while shopping, and spending extra time looking for product information. These issues make shopping slow and stressful. At the same time, shop owners want to make the shopping process faster and reduce the number of staff needed for billing and support. To solve these problems, we are creating a Smart Shopping Cart using RFID (Radio Frequency Identification) technology. In this system, every product in the store will have a small RFID tag attached to it. The shopping cart will have an RFID reader, an Arduino board (small computer), and an LCD screen. When a customer puts an item into the cart, the RFID reader will automatically read the tag and show the item's name, price, and total cost on the screen. If the customer removes the item from the cart, the display will update the cost and quantity again. This way, the customer always knows what they have added and how much it costs—without needing to wait in line for billing. This smart cart will help customers save time and shop more comfortably. It will also help store owners by reducing the need for extra staff and improving the overall shopping experience. With this system, shopping becomes quicker, smarter, and more convenient.*

**KEY WORDS:** Smart Shopping Cart, Shopping Cart, Smart Shopping.

## I. INTRODUCTION

Internet of Things (IOT) is a concept where each device is assigned to an IP address and through that IP address anyone makes that device identifiable on internet. The mechanical and digital machines are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.

Basically, it started as the “Internet of Computers.” Research studies have forecast an explosive growth in the number of “things” or devices that will be connected to the Internet. The resulting network is called the “Internet of Things” (IoT). The recent developments in technology which permit the use of wireless controlling environments like, Bluetooth and Wi-Fi that have enabled different devices to have capabilities of connecting with each other. Using a WIFI shield to act as a Micro web server for the Arduino which eliminates the need for wired connections between the Arduino board and computer which reduces cost and enables it to work as a standalone device. Shopping in supermarkets can be time-consuming and stressful. Customers often have to wait in long queues, worry about the total cost of their items, and get little help with product information. At the same time, store owners face high labor costs and customer dissatisfaction. With the rise of new technologies like RFID and IoT, shopping can be made smarter and faster. A smart shopping cart can help solve these problems by automatically reading items, showing their price, and updating the total cost in real time. This makes shopping more convenient for customers and reduces manual work for store staff. The scope of this project is to design and develop a Smart Shopping Cart system that enhances the in-store shopping experience using RFID technology. The system will be implemented in supermarkets, grocery stores, and retail outlets to automate item detection and billing. Each product will have an RFID tag, and the shopping cart will be equipped with an RFID reader, Arduino board, and LCD display. As customers add or remove items from the cart, the system will automatically detect the changes and display the item names, prices, and total cost in real time.

## II. LITERATURE SURVEY

[1] A. Patel and M. Shah, "Smart Trolley System for Automated Billing Using RFID and IoT," 2020.

This paper presents a smart shopping trolley that automates the billing process using RFID tags and Internet of Things (IoT) technology. The system reduces checkout time by scanning product RFID tags as they are placed in the cart, with the cost being updated in real-time on an attached display. The authors highlight how cloud integration enables centralized billing and inventory management. The system eliminates the need for manual billing and is especially relevant for reducing queue times in high-traffic retail environments.

[2] K. Raj and S. Agarwal, "Smart Shopping Cart Using Arduino and RFID," 2019. This study introduces a prototype built using Arduino microcontroller, RFID readers, and a simple LCD display. The cart reads RFID tags and updates the total amount dynamically. The authors also include a wireless transmission of cart data to the billing counter. Despite its simplicity, the system provides a low-cost and energy-efficient solution for small retail setups. The work emphasizes affordability and reliability, making it ideal for deployment in emerging markets.

[3] R. Kumar and L. Sharma, "AI-Based Object Recognition for Smart Shopping Carts," 2021. The authors present a vision-based shopping cart system using convolutional neural networks (CNNs) for object recognition. Instead of relying on RFID, the camera mounted on the cart identifies items as they are placed inside. The system uses pretrained models like ResNet50 for item detection and classification. The paper shows that the vision-based model achieves over 92% accuracy across diverse grocery items and offers scalability in RFID-less environments.

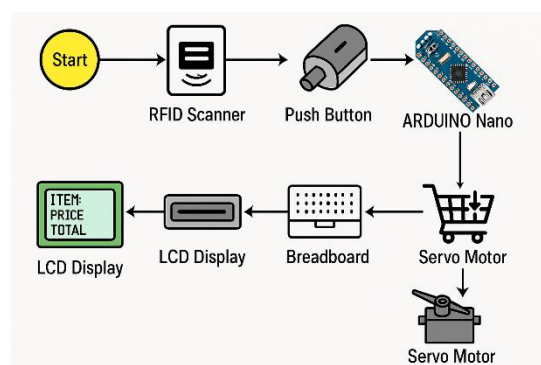
[4] S. Mehta and P. Joshi, "Smart Cart System for Automatic Billing Using Barcode Scanning and Android App," 2018. This work proposes a shopping cart equipped with barcode scanning and a Bluetooth module to send data to a customer's mobile application. The app maintains a live bill and allows online payments at checkout. The authors stress the importance of user convenience, suggesting that integrating smartphone technology increases user engagement while lowering hardware costs.

[5] N. Thakur and D. Singh, "Smart Cart with Real-Time Inventory Monitoring and Theft Prevention," 2022.

This paper explores an advanced smart cart system that combines RFID with weight sensors and a cloud backend. Real-time inventory updates help store managers maintain stock levels, while the weight sensors act as a theft prevention mechanism by validating the scanned product's weight against actual cart weight. The authors argue this hybrid system enhances security and operational efficiency.

## III. SYSTEM MODEL

The system described is a smart shopping cart implemented using an Arduino Nano microcontroller. It is designed to simplify and digitize the shopping experience using embedded electronics. The core function of the cart is to automatically identify items using RFID technology and display the total bill on an LCD screen. The system also includes a servo motor to lock or unlock the cart based on successful verification and checkout.



**Fig: ARCHITECTURE OF SMART SHOPPING CART SYSTEM USING ARDUINO NANO**

### Input Design

Input design in the smart shopping cart system is crucial as it ensures smooth interaction between hardware and users. Key inputs include RFID tags, a push button for confirmation, and cart status signals.

1. RFID-Based Item Detection: RFID tags are attached to each product. The RFID scanner reads these tags when items are placed in the cart. Each tag corresponds to specific item details and price, which are processed and displayed.

2. Push Button for Interaction: A push button allows users to confirm selections or initiate checkout. It ensures deliberate user action to avoid false reads or unintended updates.

3. **Embedded Input Wiring:** Jumper wires and breadboards are used for reliable signal transmission between the scanner, Arduino, and LCD.

### **Objectives of Output Design**

Output design focuses on providing clear, real-time feedback to users for a seamless experience.

1. **LCD Display:** Shows item names, prices, and running totals. Displays messages like “Scan Item,” “Item Added,” or “Thank You.”

2. **Servo Motor Lock System:** Controls access to the cart. Locks until a valid checkout is completed and unlocks afterward.

3. **Error Handling and Feedback:** Displays error messages for unrecognized tags or system malfunctions. Prompts the user to retry or seek assistance.

## **IV. IMPLEMENTATION**

### **1. System Development Process**

#### **Component Integration:**

The Smart Shopping Cart system is built using hardware components such as an RFID scanner, RFID tags, push button, servo motor, LCD screen, Arduino Nano, jumper wires, and a breadboard. Each component is connected and tested on the breadboard for functionality.

#### **Hardware Setup:**

- **RFID Scanner and Tags:** Each item is tagged with an RFID tag. The scanner reads the tag and sends the data to the Arduino Nano.
- **Push Button:** Used to simulate the “add to cart” function or confirm item addition after scanning.
- **Servo Motor:** Controls a small gate mechanism to allow or restrict access to the cart or item shelf.
- **LCD Display:** Provides real-time feedback to the user such as item name, price, and running total.

#### **Software Logic Development:**

- **Microcontroller Programming:** The Arduino Nano is programmed using the Arduino IDE. The code handles tag scanning, push button logic, servo control, and LCD updates.

- **Workflow:**

1. User scans an item using the RFID scanner.
2. Presses the push button to confirm the addition.
3. Arduino Nano verifies the tag and updates the total.
4. Item information is displayed on the LCD.
5. Servo motor opens gate if the item is accepted.

#### **Error Handling and Optimization:**

- Invalid tags are flagged with a message like “Invalid Item.”
- Duplicate scans are prevented by tracking previously scanned tags.
- System ensures real-time processing for user convenience.

### **2. User Interaction with the System**

#### **Provide Input:**

Users interact with the cart by scanning items and pressing the push button to confirm. The system ensures consistent input processing and handles possible delays or repeated actions.

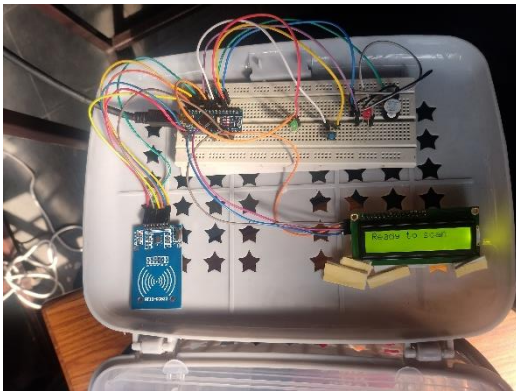
#### **View Output:**

The LCD display shows:

- Item name and price.
- Total cost so far.
- Final message (e.g., “Thank You” after shopping is complete).

This user-friendly setup allows seamless interaction while ensuring each scanned item is properly logged and processed.

## V. RESULTS



**Fig 1. Smart Shopping Cart**

## VI. CONCLUSION

The Smart Shopping Cart using Arduino Nano effectively demonstrates how automation and embedded systems can revolutionize the traditional shopping experience. By integrating components such as an RFID scanner, LCD display, servo motor, and push button, this project presents a functional prototype capable of real-time product identification, price calculation, and checkout assistance. The Arduino Nano serves as the central controller, ensuring accurate data processing and smooth interaction between the hardware components. The system successfully detects products using RFID tags as they are placed in the cart, displaying item information and running totals on the LCD via the I2C interface. The push button allows the user to confirm or reset their cart, while the servo motor can be used to trigger a lock or alert mechanism to enhance cart security and prevent theft or item mismatch. The use of basic yet effective components like jumper wires and a breadboard makes the system modular, cost effective, and ideal for prototyping or educational use.

## VII. FUTURE ENHANCEMENT

To improve the functionality, user experience, and scalability of the Smart Shopping Cart system, several enhancements can be implemented. One of the key upgrades would be the integration of wireless communication modules such as Bluetooth, Wi-Fi, or GSM to enable real-time syncing of product data with a central store server or cloud database. This would allow seamless inventory management, digital receipts, and remote monitoring of carts. Developing a dedicated mobile application could significantly enhance user interaction by

displaying detailed billing information, offering personalized promotions, and allowing users to track their shopping list in real-time. Implementing automatic payment features using NFC or QR code-based mobile payments could eliminate the need for traditional checkouts, further streamlining the shopping process. For improved item detection accuracy, barcode or QR code scanning modules could be added alongside RFID, enabling support for a wider range of product tagging systems. Load cells or weight sensors could also be integrated to verify item quantities and prevent billing errors or theft. The addition of GPS modules would enable cart tracking within large stores, helping users locate items or navigate the store more efficiently. To increase energy efficiency and make the system eco-friendly, solar-powered modules could be integrated for outdoor market carts. Additionally, voice feedback or multilingual display support could be added to assist users with visual impairments or language barriers.

## VIII. REFERENCES

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